TruthVoice — how the website works

Team guide · RMIT UNLOCKED hackathon (Challenge 2, Tide) · 22 Oct 2026

1. The idea in one minute

Students hold back honest course feedback because the people who run the survey system (IT admins, developers, database owners) can read every comment and could work out who wrote it. TruthVoice keeps every comment **encrypted**, so the database, the IT admin and the lecturer only ever see scrambled text.

Feedback is still useful: an **AI summariser** can read one course's comments, but only under a **lease**: a short, expiring permission that **2 of 3 independent approvers** (student representative, dean, quality-assurance officer) must grant together. The lecturer receives only the AI's **theme summary** (e.g. "assessment (4), materials (1)"), never the students' sentences. When the lease expires, the AI is locked out again.

This combines our team's TruthVoice topic (anonymous RMIT course evaluation) with the "Leash" engine (AI access only under a quorum-signed, expiring lease).

2. Who is who

Role	What they can do
Student	Writes feedback. It is encrypted on submit; no student id is stored.
University IT admin	Runs the database and server. Sees only ciphertext. Cannot read or approve.
Lecturer	Asks for an AI summary of their course. Never sees raw comments. Cannot approve their own request.
Approvers (student rep, dean, QA officer)	Any 2 of the 3 must approve before the AI may read a course.
AI summariser	Reads one course's comments only while a lease is active; outputs themes and counts only. Cannot approve anything.

3. How to open it

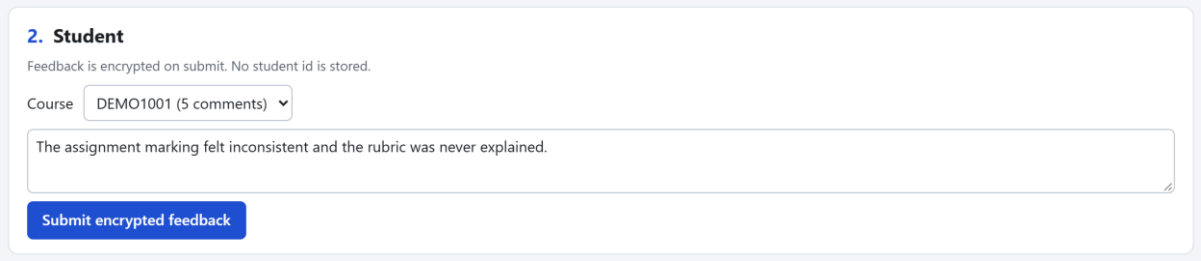
- From the app folder run **npm install** then **npm run dev**, and open **http://localhost:3000/truthvoice**.
- The yellow badge says **"Demo mode — Tide network simulated"**: the screens below run without TideCloak, using a mock of the Tide network (see section 5).
- All data is synthetic: 3 fictional courses (DEMO1001–DEMO3003) with 15 seeded comments. Restart the server to reset.
- The page is one long screen with numbered parts. The walkthrough below follows the demo story in order.

4. Walkthrough of the page



Header: the product name, the one-line pitch and the mode badge

1. Student — submit feedback



Part 2: the student picks a course and writes feedback

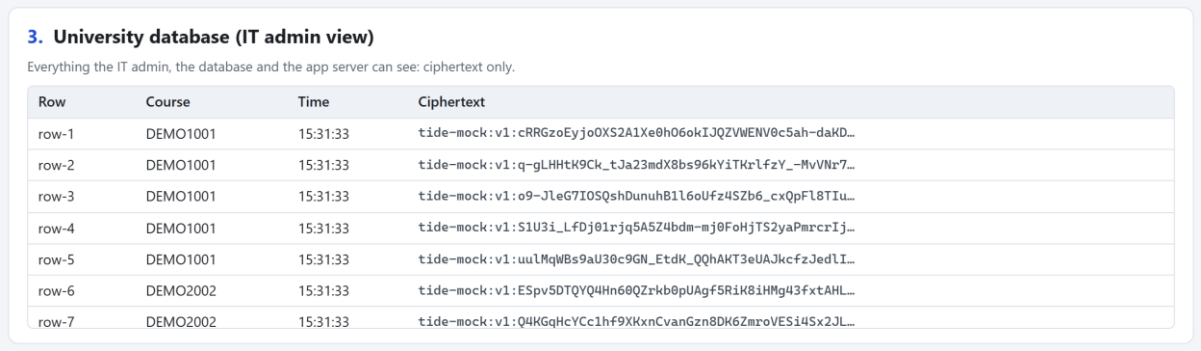
What you see

- A course dropdown (with how many comments it already has) and a text box (max 1000 characters).
- "Submit encrypted feedback" encrypts the comment before it is stored. The page never stores who wrote it.

Why it matters

Students can write honestly because nobody who runs the system can read or trace the comment.

2. University database (IT admin view)



Part 3: what the IT admin, the database and the server can see

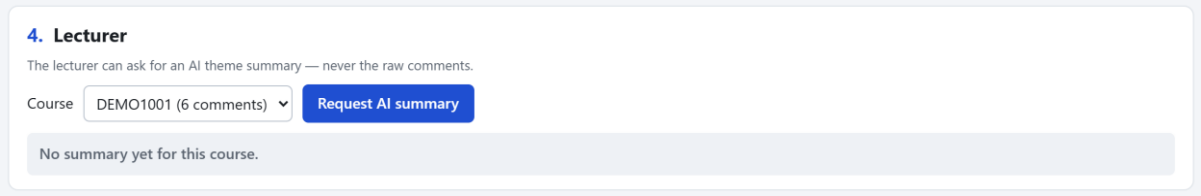
What you see

- Every stored row: row id, course, time and the ciphertext (scrambled text, shortened).
- There is no plaintext anywhere in this table, including the comment just submitted.

Why it matters

This is the core trust claim: even full database and server access gives an admin nothing readable.

3. Lecturer — request an AI summary



Part 4: the lecturer asks for a summary; nothing is shown until a lease exists

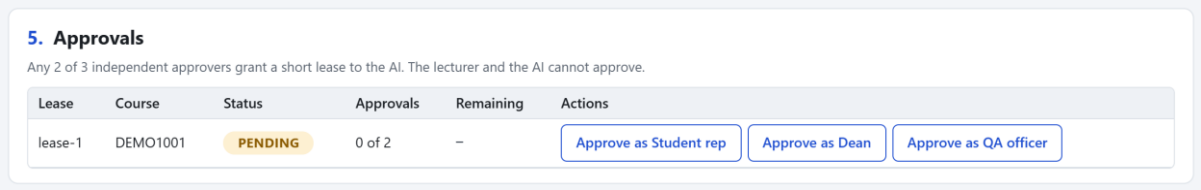
What you see

- "Request AI summary" opens a lease request for the chosen course (it appears in Part 5 as PENDING).
- The result box shows "No summary yet" until the AI has been allowed to read the course.

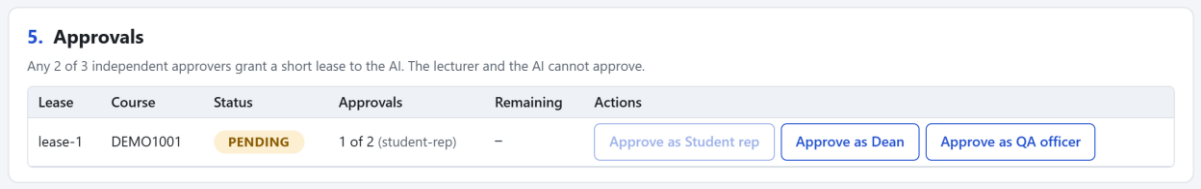
Why it matters

The person who wants the information cannot grant it to themselves.

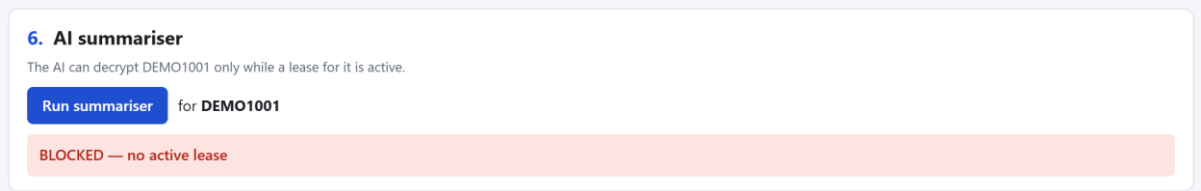
4. Approvals — before quorum



Part 5: the new lease request, waiting for approvers



After the student rep approves: still PENDING, 1 of 2



Part 6: with only 1 approval, the AI is still BLOCKED

What you see

- Each lease shows its course, status, approvals so far and time remaining.
- Buttons "Approve as Student rep / Dean / QA officer"; a button greys out once that person has approved.
- One approval is not enough: running the AI summariser still gives "BLOCKED — no active lease".

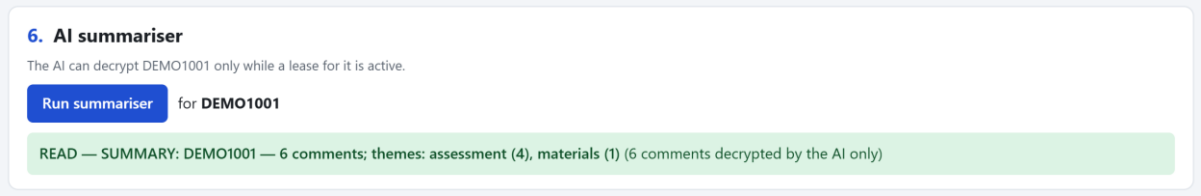
Why it matters

No single person (admin, dean or anyone) can open the data alone.

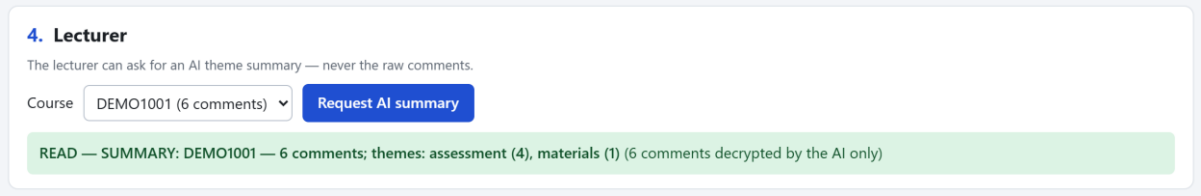
5. Approvals — quorum reached, AI reads themes only



Part 5: after the dean approves, the lease is ACTIVE (2 of 2) with a 15-minute countdown



Part 6: the AI decrypts only DEMO1001 and returns themes and counts



Part 4: the lecturer receives the same theme summary, never the raw sentences

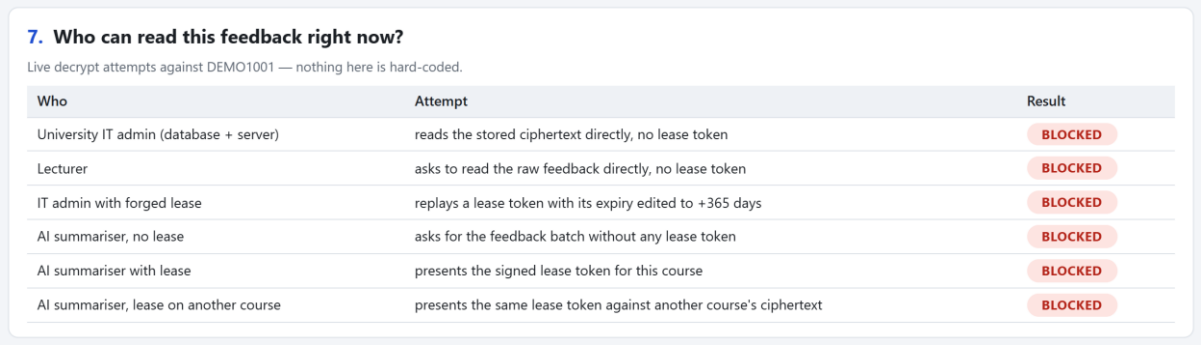
What you see

- The lease becomes ACTIVE and shows the remaining time (about 15:00).
- "Run summariser" now returns READ with a summary like "6 comments; themes: assessment (4), materials (1)".
- The summary is built from keyword themes and never repeats any student's wording (a test checks no 4 consecutive words of a comment ever appear).

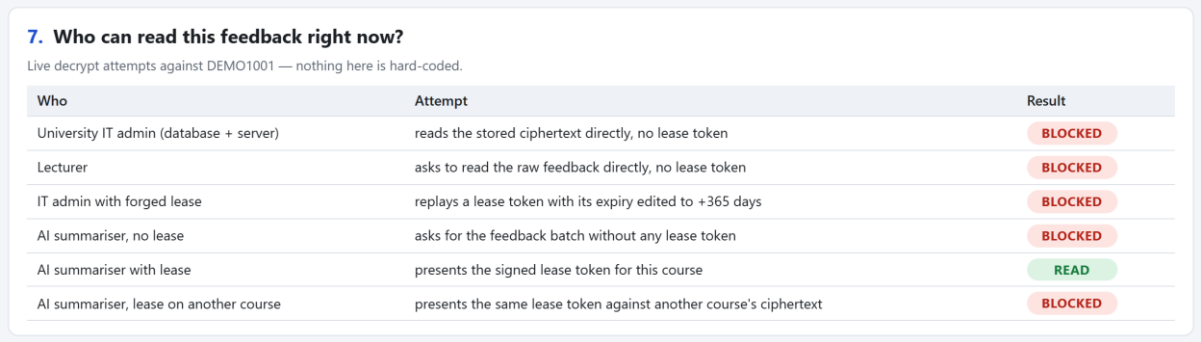
Why it matters

Lecturers get actionable themes; students' sentences, which could identify them, never leave the encrypted store.

6. Who can read this feedback right now?



Part 7 before quorum: every attempt is BLOCKED



Part 7 while the lease is active: only "AI summariser with lease" can READ

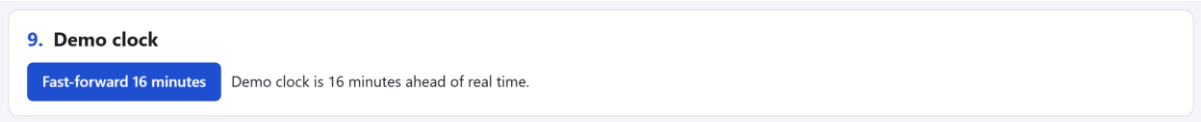
What you see

- Six real decrypt attempts run live against DEMO1001; nothing is hard-coded.
- IT admin, lecturer, a forged lease (expiry edited to +365 days), the AI without a lease, and the AI using its lease on a different course are always BLOCKED.
- Only the AI, holding the signed lease for this course, can READ, and only while the lease is active.

Why it matters

This table is the easiest thing for judges to watch: it proves who can and cannot read, at every moment of the story.

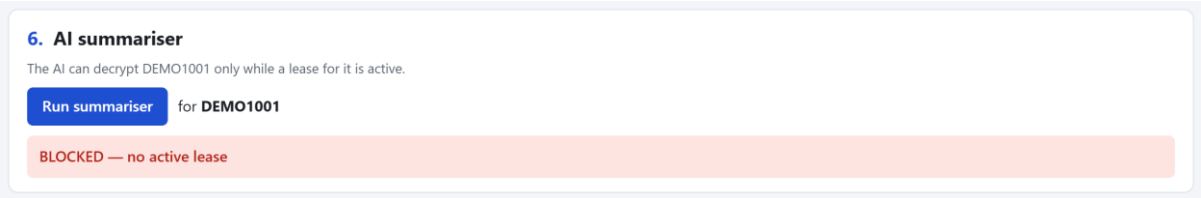
7. Demo clock — the lease expires



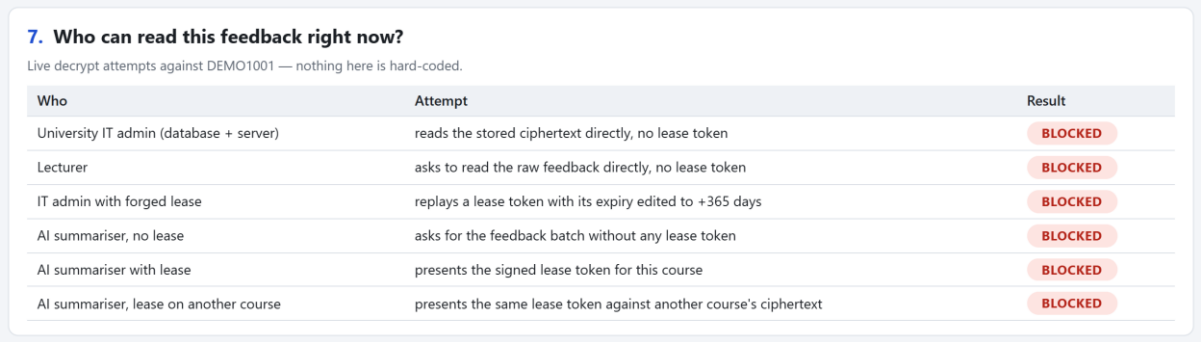
Part 9: fast-forward the demo clock by 16 minutes



Part 5: the lease is now EXPIRED



Part 6: the AI is BLOCKED again



7. Who can read this feedback right now?

Live decrypt attempts against DEMO1001 — nothing here is hard-coded.

Who	Attempt	Result
University IT admin (database + server)	reads the stored ciphertext directly, no lease token	BLOCKED
Lecturer	asks to read the raw feedback directly, no lease token	BLOCKED
IT admin with forged lease	replays a lease token with its expiry edited to +365 days	BLOCKED
AI summariser, no lease	asks for the feedback batch without any lease token	BLOCKED
AI summariser with lease	presents the signed lease token for this course	BLOCKED
AI summariser, lease on another course	presents the same lease token against another course's ciphertext	BLOCKED

Part 7: every row is BLOCKED again

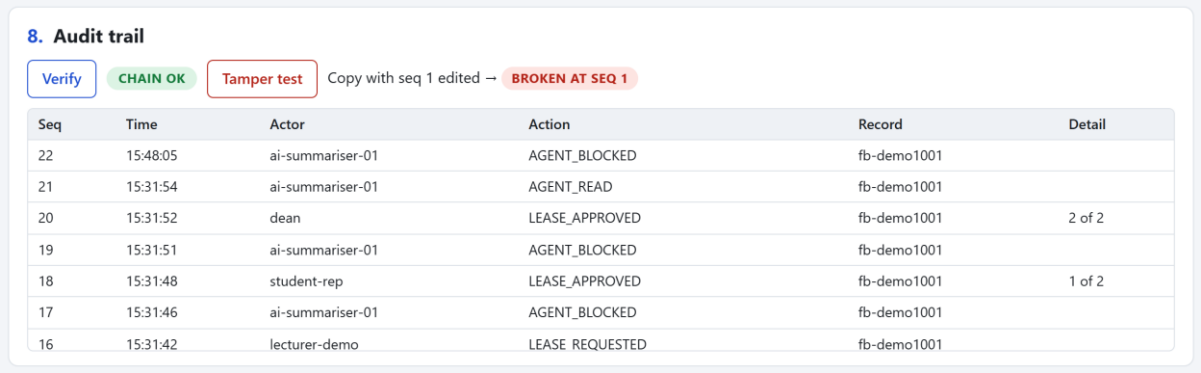
What you see

- "Fast-forward 16 minutes" moves the demo clock past the 15-minute lease.
- The lease turns EXPIRED, the AI is blocked again, and the access table goes back to all BLOCKED.

Why it matters

Access is temporary by design: nobody has to remember to revoke it.

8. Audit trail



8. Audit trail

Verify | CHAIN OK | Tamper test | Copy with seq 1 edited → | BROKEN AT SEQ 1

Seq	Time	Actor	Action	Record	Detail
22	15:48:05	ai-summariser-01	AGENT_BLOCKED	fb-demo1001	
21	15:31:54	ai-summariser-01	AGENT_READ	fb-demo1001	
20	15:31:52	dean	LEASE_APPROVED	fb-demo1001	2 of 2
19	15:31:51	ai-summariser-01	AGENT_BLOCKED	fb-demo1001	
18	15:31:48	student-rep	LEASE_APPROVED	fb-demo1001	1 of 2
17	15:31:46	ai-summariser-01	AGENT_BLOCKED	fb-demo1001	
16	15:31:42	lecturer-demo	LEASE REQUESTED	fb-demo1001	

Part 8: every action is logged in a tamper-evident chain

What you see

- Every request, approval, AI read and AI block is logged with time, actor and course (never comment text).
- "Verify" checks the hash chain: CHAIN OK.
- "Tamper test" edits one entry in a copy of the log; verification immediately reports BROKEN AT SEQ 1.

Why it matters

Quality teams can prove afterwards who accessed what, and that nobody quietly edited the record.

5. Demo mode vs Tide mode (be precise when presenting)

	Demo mode (shown above)	Tide mode (built, live test pending)
Encryption	Simulated Tide network inside our server (mock gateway)	Real Tide: encrypted in the student's browser; keys held by Tide's decentralised network; our server never decrypts
Sign-in	None	Real TideCloak sign-in with DPoP (token bound to the device)
2-of-3 approvals	Enforced by our app	Enforced by our app
Who / which course / expiry of the AI lease	Enforced by the mock gateway	Designed to be enforced by the Tide network (built-in RelationshipAttestation:1 contract)
Status	Working; tested end to end	Code builds and passes unit tests; API rejects unauthenticated calls (401). Live run blocked by Tide-side issues (below)

Honest limits: in Tide mode the 2-of-3 rule is checked by our app, while Tide is designed to enforce who can read, which course and until when. Tide verifies the AI operator's account, not the AI code behind it. Please do not claim the live Tide checks have run until they have.

6. Where real Tide stands today

- **Licence:** TideCloak 0.14.41 (the newest image) cannot activate a free licence; 0.14.40 works. We pinned 0.14.40 and our realm "truthvoice" is licensed.
- **Browser:** Comet (and extension-heavy browsers) block Tide's sign-up page from reaching our local server. Use a clean Chrome profile.
- **Account creation:** fails inside Tide's network (server ork7 returns 500 on the final step; ork9 does not respond). A bug report is ready in TIDE-SUPPORT-REQUEST.md for the Tide Discord.
- Until Tide fixes this, the video shows demo mode, clearly labelled "Tide network simulated".

7. Words you will hear

Term	Meaning
Ciphertext	Encrypted, unreadable text. What the database stores instead of the comment.
Lease	A short, expiring permission for the AI to read one course's feedback.
Quorum (2 of 3)	The minimum number of independent approvers needed to grant a lease.

Term	Meaning
TideCloak	Tide's sign-in server (based on Keycloak) that connects the app to the Tide network.
ORK	One of Tide's independent key servers. Keys are split across many ORKs, so no single server or admin holds them.
DPoP	Proof that a sign-in token is being used from the device it was issued to.
Audit chain	Log entries linked by hashes, so editing any entry is detected.

8. Where things are in the repo

Path	What it is
README.md	Submission overview (≤ 300 words), setup and tests
VIDEO-SCRIPT.md	2-minute recording script with timings and voice-over
core/	The engine (encryption gateway, leases, approvals, AI summariser, audit) with 389 tests
app/app/truthvoice/	This web page (demo and Tide versions)
runs/	Test records, planted-bug checks and the Tide licence diagnosis